

CONCEPT 28.6

Protists play key roles in ecological communities (pp. 614–615)

- Protists form mutualistic and parasitic relationships that affect their symbiotic partners and many other members of the community.
- Photosynthetic protists are among the most important **producers** in aquatic communities. Because they are at the base of the food web, factors that affect photosynthetic protists affect many other species in the community.

? Describe several protists that are ecologically important.

TEST YOUR UNDERSTANDING

➔ For more multiple-choice questions, go to the **Practice Test** in the **Mastering Biology** eText or Study Area, or go to goo.gl/GruWRg.

Levels 1-2: Remembering/Understanding

1. Plastids that are surrounded by more than two membranes are evidence of
(A) evolution from mitochondria.
(B) fusion of plastids.
(C) origin of the plastids from archaea.
(D) secondary endosymbiosis.
2. Biologists think that endosymbiosis gave rise to mitochondria before plastids partly because
(A) the products of photosynthesis could not be metabolized without mitochondrial enzymes.
(B) all eukaryotes have mitochondria (or their remnants), whereas many eukaryotes do not have plastids.
(C) mitochondrial DNA is less similar to prokaryotic DNA than is plastid DNA.
(D) without mitochondrial CO₂ production, photosynthesis could not occur.
3. Which group is correctly paired with its description?
(A) diatoms—important consumers in aquatic communities
(B) diplomonads—protists with modified mitochondria
(C) apicomplexans—producers with intricate life cycles
(D) red algae—acquired plastids by secondary endosymbiosis
4. According to the phylogeny presented in this chapter, which protists are in the same eukaryotic supergroup as plants?
(A) green algae
(B) dinoflagellates
(C) red algae
(D) both A and C
5. In a life cycle with alternation of generations, multicellular haploid forms alternate with
(A) unicellular haploid forms.
(B) unicellular diploid forms.
(C) multicellular haploid forms.
(D) multicellular diploid forms.

Levels 3-4: Applying/Analyzing

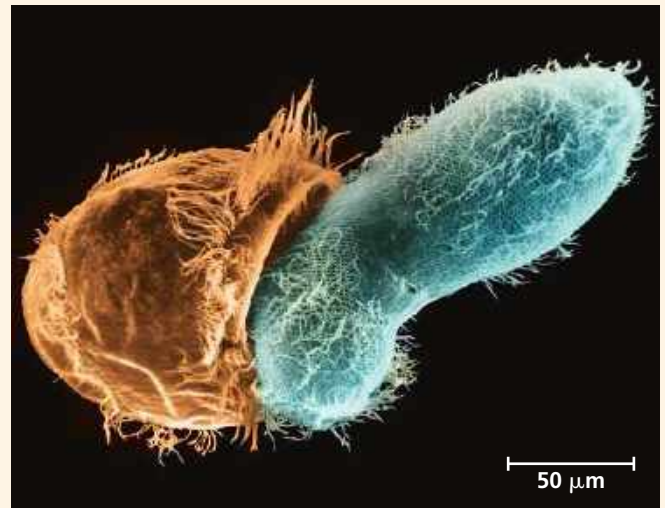
6. Based on the phylogenetic tree in Figure 28.5, which of the following statements is correct?
(A) Excavata and SAR form a sister group.
(B) The most recent common ancestor of SAR is older than that of Unikonta.
(C) The most basal (first to diverge) eukaryotic supergroup cannot be determined.
(D) Excavata is the most basal eukaryotic supergroup.

7. **EVOLUTION CONNECTION • DRAW IT** Medical researchers seek to develop drugs that can kill or restrict the growth of human pathogens yet have few harmful effects on patients. These drugs often work by disrupting the metabolism of the pathogen or by targeting its structural features.

Draw and label a phylogenetic tree that includes an ancestral prokaryote and the following groups of organisms: Excavata, SAR, Archaeplastida, Unikonta, and, within Unikonta, amoebozoans, animals, choanoflagellates, fungi, and nucleariids. Based on this tree, hypothesize whether it would be most difficult to develop drugs to combat human pathogens that are prokaryotes, protists, animals, or fungi. (You do not need to consider the evolution of drug resistance by the pathogen.)

Levels 5-6: Evaluating/Creating

8. **SCIENTIFIC INQUIRY** Applying the “If . . . then” logic of deductive reasoning (see Concept 1.3), what are a few of the predictions that arise from the hypothesis that plants evolved from green algae? Put another way, how could you test this hypothesis?
9. **WRITE ABOUT A THEME: INTERACTIONS** Organisms interact with each other and the physical environment. In a short essay (100–150 words), explain how the response of diatom populations to a drop in nutrient availability can affect both other organisms and aspects of the physical environment (such as carbon dioxide concentrations).
10. **SYNTHESIZE YOUR KNOWLEDGE**



This micrograph shows a single-celled eukaryote, the ciliate *Didinium* (left), about to engulf its *Paramecium* prey, which is also a ciliate. Identify the eukaryotic supergroup to which ciliates belong and describe the role of endosymbiosis in the evolutionary history of that supergroup. Are these ciliates more closely related to all other protists than they are to plants, fungi, or animals? Explain.

For selected answers, see Appendix A.